DATA CLEANING REPORT

The employee dataset consisted of 1470 rows and 36 columns. This report documents the data-cleaning steps performed, including the columns removed, duplicate records, missing/null values, and other data-quality checks conducted on the dataset.

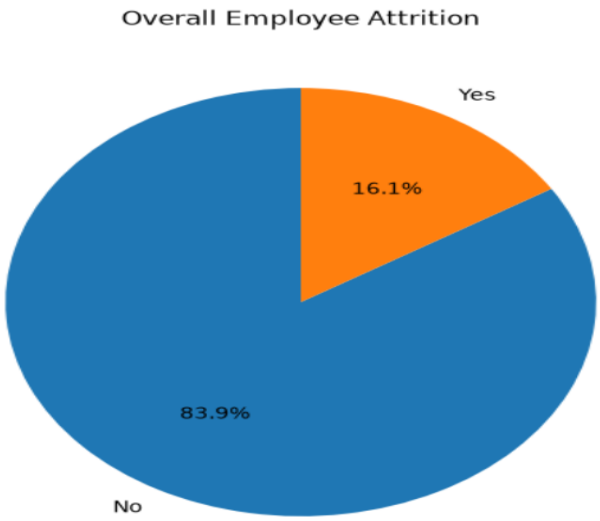
CHECK	FINDING	ACTION
Missing values	No missing values found	No action
Duplicate rows	No duplicates found	No action
Data types	Checked and appropriate	No action
Categorical values	No inconsistencies found	No action
Numerical values	No invalid values found	No action
Constant values	EmployeeCount,Over18,StandardHours Had constant value “1”	Removed
CurrManager	Completely empty	Removed

The Employee dataset was checked for missing values, duplicate rows and values, incorrect data types, inconsistent categorical values, invalid numerical values and constant values. No major data quality issues found. However the columns *EmployeeCount,Over18,StandardHours* were removed due to their constant value “1” and “CurrManager” column was removed from the working Employee dataset due to its completely empty column.

The existing count of rows and columns after data cleaning process- 1470 X 33

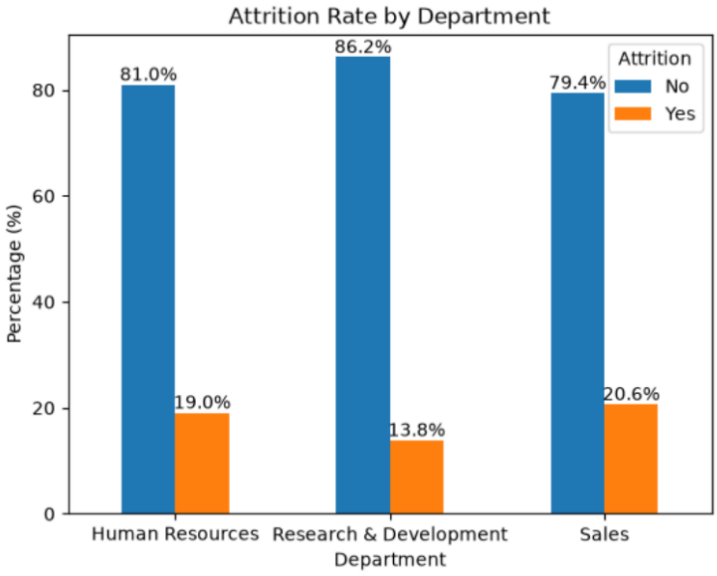
EDA REPORT

2.1 TOTAL ATTRITION RATE



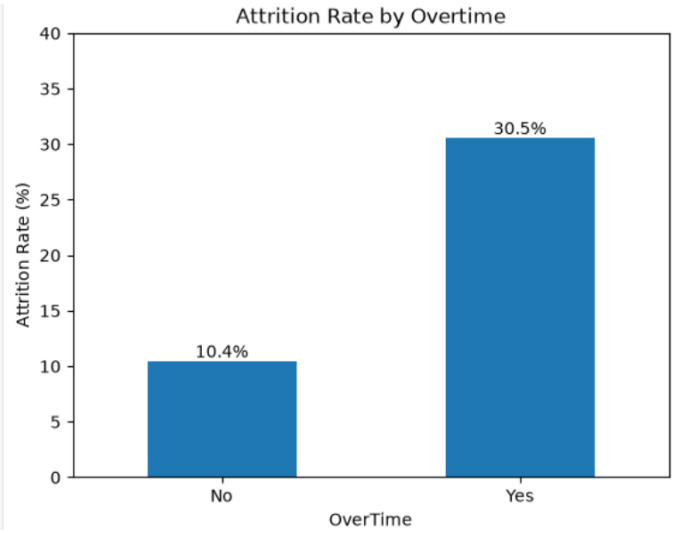
The dataset consists of 1470 employees out of which 237 employees i.e. 16.12% left the company while 1233 employees i.e. 83.88% remained. Hence, the overall employee attrition rate in the dataset is 16.12%.

2.2 EMPLOYEE ATTRITION BY DEPARTMENT



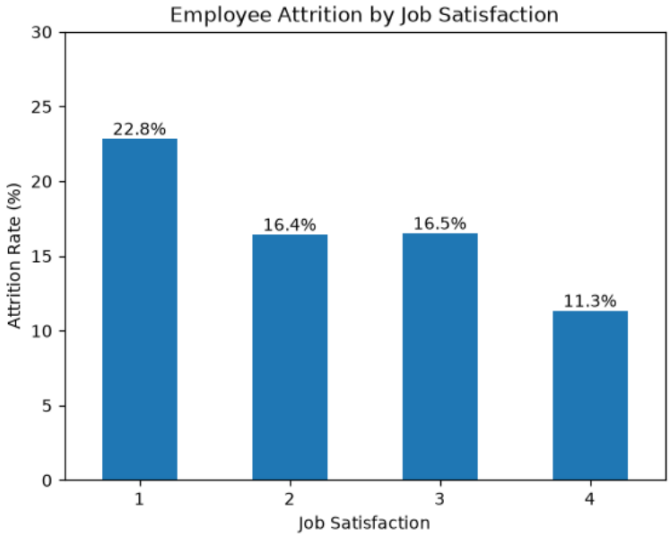
Attrition rates were compared across the three departments. The Sales department recorded an attrition rate of 20.63%, followed by Human Resources at 19.05%. Research & Development had the lowest attrition rate at 13.84%. This indicates that attrition varies across departments, with a relatively higher proportion of employees leaving from Sales compared with the other departments.

2.3 EMPLOYEE ATTRITION BASED ON OVERTIME



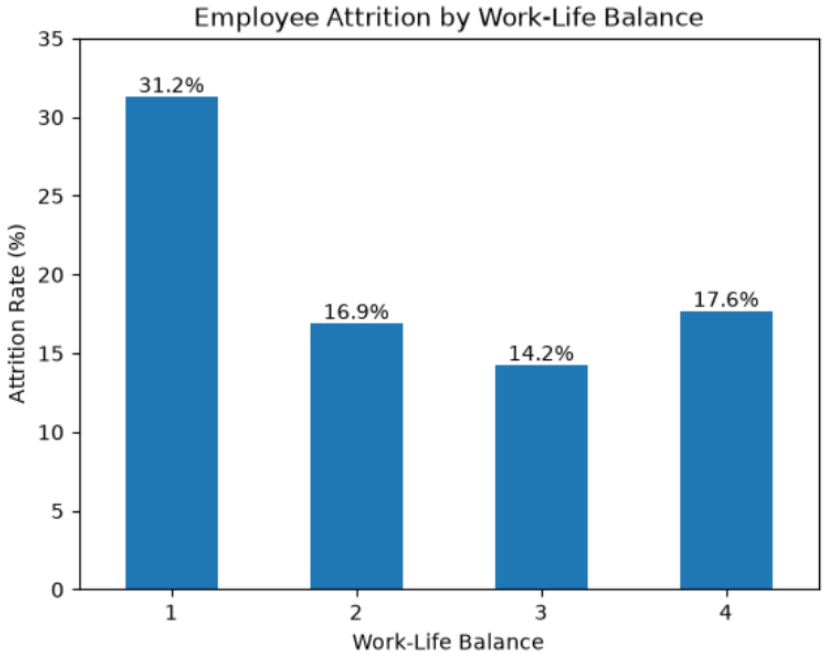
The analysis shows a difference in attrition rates based on overtime status. Employees who worked overtime had an attrition rate of 30.53%, compared with 10.44% among employees who did not work overtime. This indicates that overtime status is associated with a higher attrition rate in this dataset.

3.4 EMPLOYEE ATTRITION BASED ON JOB SATISFACTION



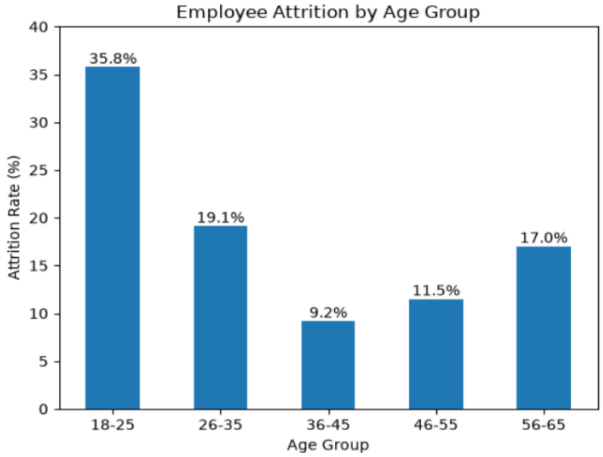
Employees with the lowest job satisfaction level i.e. 1 had the highest attrition rate at 22.84%, while those with the highest satisfaction level i.e. 4 had a much lower attrition rate of 11.33%. This suggests that employees who are less satisfied with their jobs may be more likely to leave the company.

3.5 EMPLOYEE ATTRITION BASED ON WORK LIFE BALANCE



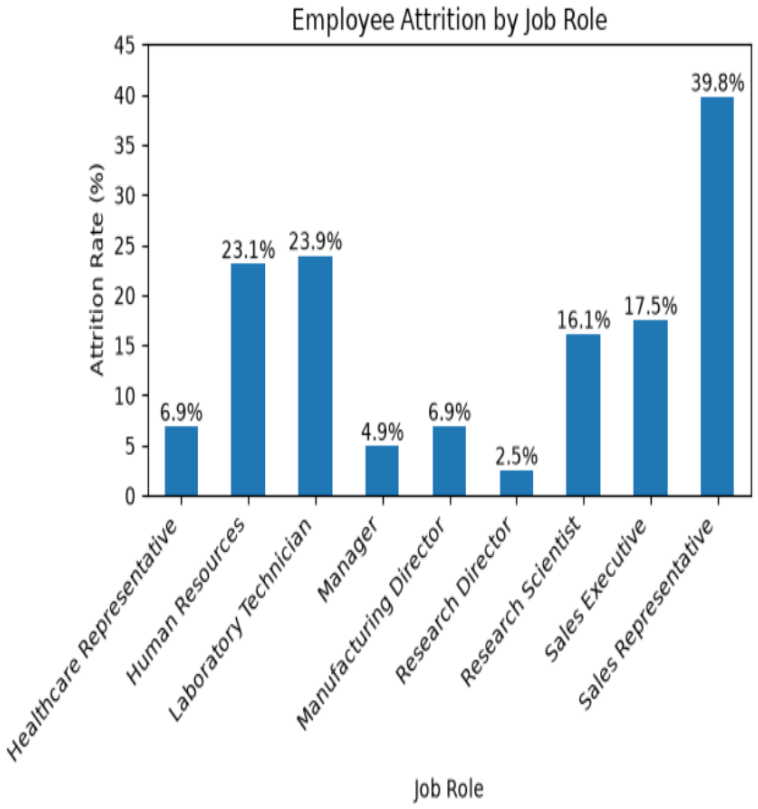
Employees with the lowest work-life balance level i.e. 1 had the highest attrition rate at 31.25%. The attrition rate was lower for levels 2 and 3 at 16.86% and 14.22% respectively. At level 4, the rate increased slightly to 17.65%. Overall, employees with a very low work-life balance showed a noticeably higher attrition rate in this dataset.

3.6 EMPLOYEE ATTRITION BY AGE GROUP



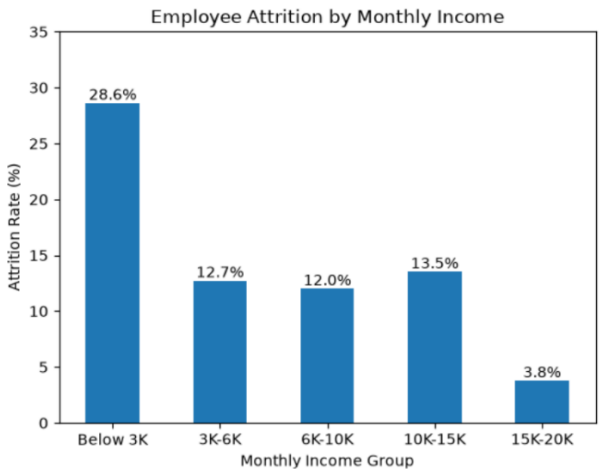
Employees aged 18–25 had the highest attrition rate at 35.77%, while the 36–45 age group had the lowest at 9.19%. The 26–35 and 46–55 age groups had attrition rates of 19.14% and 11.50%, respectively, while employees aged 56–65 had a rate of 17.02%. Overall, the data shows that younger employees, especially those aged 18–25, were more likely to leave the organization.

3.7 EMPLOYEE ATTRITION BY JOB ROLE



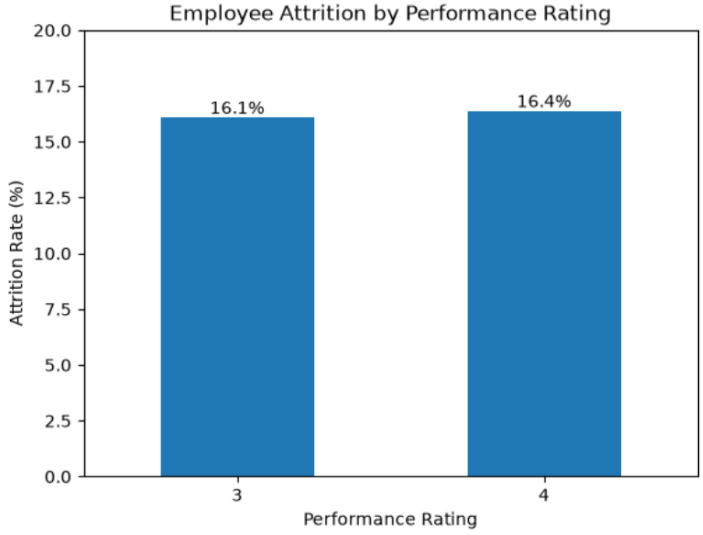
The attrition rate was quite different across the various job roles. Sales Representatives had the highest attrition rate at 39.76%, followed by Laboratory Technicians at 23.94% and Human Resources employees at 23.08%. In comparison, Research Directors had the lowest attrition rate at 2.50%, followed by Managers at 4.90%. Overall, the data shows noticeable differences in attrition across job roles, with Sales Representatives experiencing a particularly high rate of employee turnover.

3.8 EMPLOYEE ATTRITION BY MONTHLY INCOME



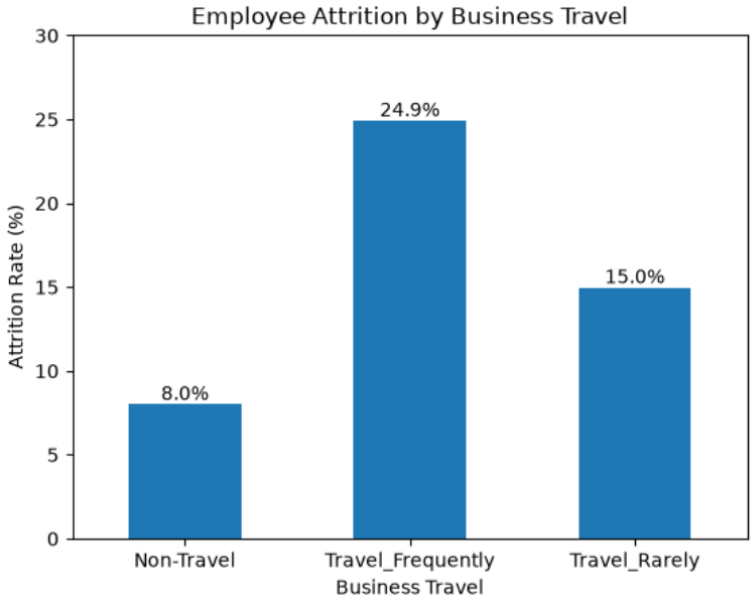
Employees in the Below 3K income group had the highest attrition rate at 28.61%. The rate was much lower for employees earning between 3K and 15K, where it stayed around 12–14%. Employees in the 15K–20K income group had the lowest attrition rate at 3.76%. Overall, the data shows that employees in the lower income group had a noticeably higher attrition rate than those in the higher income groups.

3.9 EMPLOYEE ATTRITION BY PERFORMANCE RATING



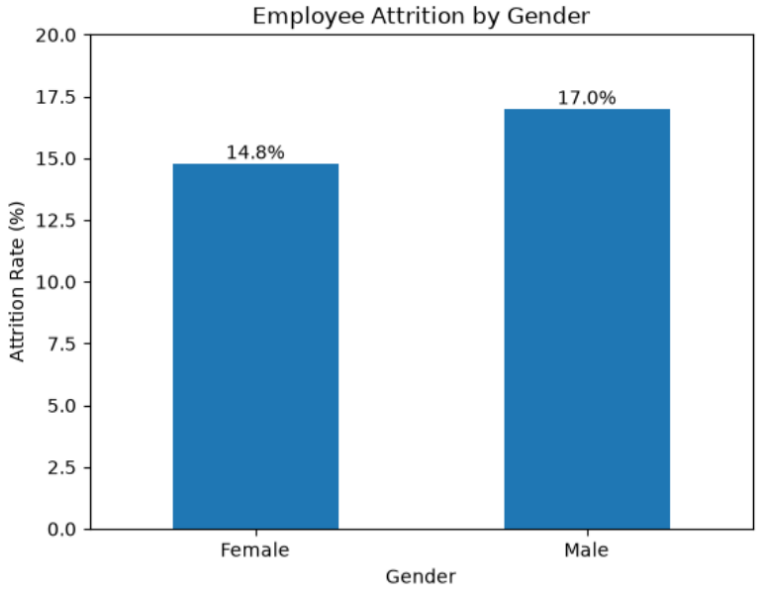
The attrition rates for the two performance ratings were very similar. Employees with a performance rating of 3 had an attrition rate of 16.08%, while those with a rating of 4 had a slightly higher rate of 16.37%. The difference is very small, suggesting that performance rating did not show much variation in attrition rates in this dataset.

3.10 EMPLOYEE ATTRITION BY BUSINESS TRAVEL



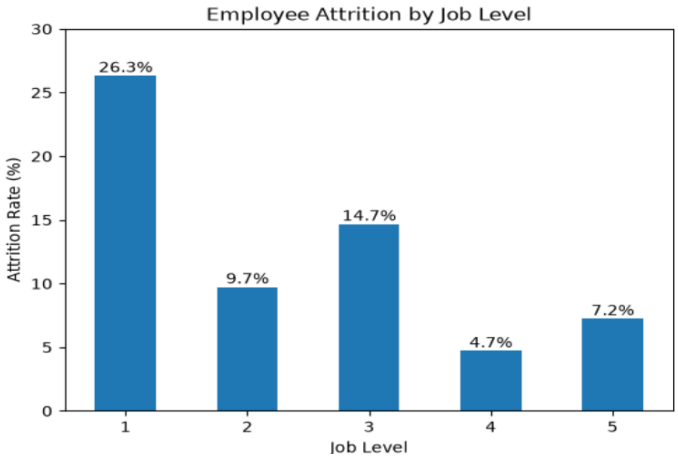
Employees who travelled frequently had the highest attrition rate at 24.91%. This was followed by employees who travelled rarely, with an attrition rate of 14.96%. Employees who did not travel had the lowest attrition rate at 8.00%. Overall, the data shows that employees who travelled more frequently had higher attrition rates in this dataset.

3.11 EMPLOYEE ATTRITION BY GENDER



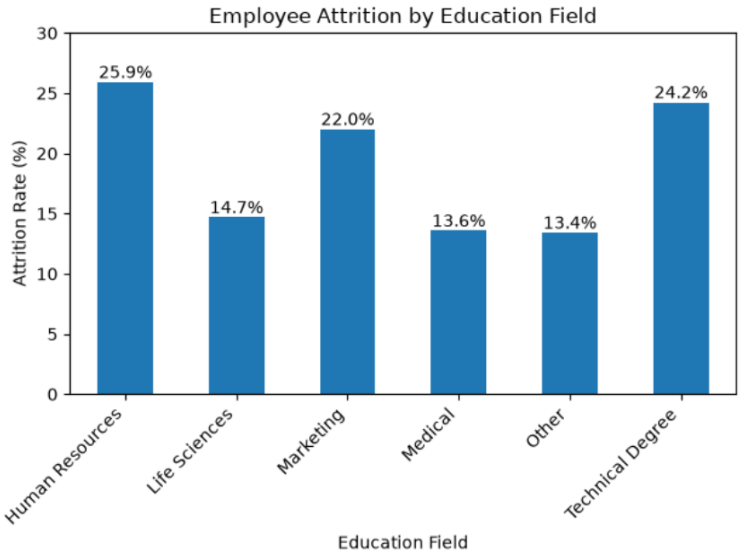
The attrition rates for male and female employees were fairly similar. Female employees had an attrition rate of 14.80%, while male employees had a slightly higher rate of 17.01%. Overall, the difference between the two groups was relatively small in this dataset.

3.12 EMPLOYEE ATTRITION BY JOB LEVEL



Employees at Job Level 1 had the highest attrition rate at 26.34%, while employees at Job Level 4 had the lowest rate at 4.72%. Job Levels 2 and 5 also had relatively lower attrition rates of 9.74% and 7.25%, respectively. Overall, the data shows noticeable differences in attrition across job levels, with employees at the entry-level showing a higher attrition rate

3.13 EMPLOYEE ATTRITION BY EDUCATION FIELD



Employees from the Human Resources education field had the highest attrition rate at 25.93%, followed by Technical Degree at 24.24% and Marketing at 22.01%. In comparison, employees from Medical, Other, and Life Sciences backgrounds had lower attrition rates. This shows that attrition varied across different education fields in the dataset.

HR KPI'S

Key HR performance indicators were calculated to provide an overall view of the workforce and employee attrition.

KPI	VALUE
Total employees	1470
Employees who left	237
Employees who stayed	1233
Attrition rate	16.12%
Retention rate	83.88%
Average monthly income	6502.93
Average age	36.92
Average years at company	7.01
Average job satisfaction	2.73
Average work life balance	2.76
Overtime rate	28.3%